

Chronic effects of a high-fat diet enriched with virgin olive oil and a low-fat diet enriched with alpha-linolenic acid on postprandial endothelial...

Traditional cardiovascular risk factors are associated with endothelial dysfunction. The vascular endothelium plays a key role in local vascular tone regulation and can be modulated by dietary fat. We propose to determine the chronic effect of three diets with different fat compositions on postprandial endothelial function and inflammatory biomarkers. Twenty healthy men followed three 4-week diets in a randomised cross-over design: a Western diet, rich in saturated fat (22% SFA, 12% MUFA and 0.4% alpha-linolenic acid (ALA), all fractions are % of energy); a Mediterranean diet, rich in MUFA (< 10 % SFA, 24 % MUFA and 0.4% ALA); a low-fat diet enriched in ALA (< 10% SFA, 12% MUFA and 2% ALA). At the end of each dietary period all subjects underwent a postprandial study. Plasma concentrations of lipid parameters, soluble intercellular cell-adhesion molecule-1, soluble vascular cell-adhesion molecule-1 (sVCAM-1), nitrates and nitrites (NOx) and endothelial function studied by laser Doppler were examined at 0, 2, 4, 6 and 8 h. The endothelium-dependent vasodilatory response was greater 4 h after the ingestion of the MUFA-rich diet than after the SFA or ALA low-fat diets ($P = 0.031$). The 4 h postprandial plasma sVCAM-1 levels were lower after the MUFA meals than after the ALA low-fat diet ($P = 0.043$). The bioavailability of NOx was higher following the MUFA diet than after the SFA and ALA low-fat diets ($P = 0.027$). We found no differences in the other parameters measured. Chronic ingestion of a Mediterranean diet avoids the postprandial deterioration of endothelial function associated with Westernised diets in healthy individuals.